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Distributed snapshots: determining global states of distributed system

The paper presents a solution in defining and recording a global state in a distributed computational system. It proposes a clear definition of identifying a global state. The algorithm records the states of processes where messages go out and states of processes where messages are received. The algorithm utilizes the messages going to the receiving processes to define the states of its incident channels.

The paper also states that a sequence of events in a distributed computation can be permuted in a way where any post-recording event preceding a pre-recording event have their orders switched. In the permuted new sequence. The state where after all pre-recording events and before all post-recording events is the global state.

I think the best part of the algorithm is that they use markers sent out from the “message-out” process to notify the “message-in” processes where exactly are the pre-recording points, therefore, help in recording the global state.

I think such definition of the global state is to seek the logical correctness of events so the global states can be reachable from any state in the computation. This is an especially useful property of the global state we are able to find because y(S*) can to extended to all states reachable by S* because of the definition of stable property.

I don’t find any shortcoming in the method proposed by the author. But I think it help if he can explain why we can detect true condition but not the false condition for the stable property at the termination of the algorithm in stability detection problem.